Learning to Feel

Perception-Action Learning, Phase-Compatible Recruitment, and Conscious Sensory Access in Self-Aware Networks

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Abstract

A physical signal can affect neural tissue without becoming a consciously accessible sensation. I propose that perception-action learning can change this relationship by altering which learned relations a sensory event recruits and how those relations influence receiving populations. In Self-Aware Networks (SAN), retained learning and present coordination jointly shape a recurrent sensory construction that guides further inference and action. The proposed contribution is a testable connection among established operations, not their invention or a completed causal demonstration assembled from separate studies.

The account distinguishes stimulus magnitude, learned significance, affective value, neural activity, receiver-relative influence and experienced intensity. It predicts acquired changes in consequential recruitment and asks whether a specified timing-dependent part contributes to conscious access. Conditional mathematics separates retuning from gain and identifies ambiguities in pooled neural measurements and sensory reports. Five constructed applications implement design checks, receiver-level inference, learned temporal selection, reciprocal statistical reconstruction and joint receiver/behavior identification. In the joint test, a receiver observation separates constrained explanations that predict identical behavior, while preserving a nonoscillatory implementation with equivalent predictions. Strong alternatives delimit their results: informative replay reproduces active learning; a nonoscillatory receiver performs at least as well as the phase-feature learner; and direct inference reproduces recurrent reconstruction. Targeted observation improves consequence prediction, not reconstruction of every represented variable.

A prospective three-arm experiment combines matched tactile probes, receiver-specific intervention and complementary access measures. Sensory discrimination, felt magnitude and decision strategy are not interchangeable outcomes. The broader SAN claim that an appropriately organized self-world learning process constitutes experience remains explicit and requires further explanation and evidence. No biological experiment or conscious artificial system is reported.

Keywords: sensory learning; conscious access; phase-dependent communication; perception-action; effective connectivity; Self-Aware Networks

1. The problem: becoming able to feel a consequential difference

Consider learning to recognize a raised mark by exploring it with a fingertip. Before learning, contact may already change receptor activity and evoke a cortical response. After learning, the same calibrated contact may identify a particular surface, anticipate an outcome, or suggest an action. These achievements are related but not identical. A system can detect a difference without knowing what it predicts; it can use a prediction without reporting the cue; and it can report a sensation more frequently without becoming more sensitive to it.

The question is whether perception-action learning can change the sensory pattern's participation in the system, rather than merely attach a label to an otherwise unchanged representation. I propose that learning establishes relations through which a local event can recruit a wider, differentiated organization. The recruited organization then changes the conditions under which subsequent evidence is received. A sensation becomes consequential through this continuing interaction, not because a separate observer examines an internal display.

This formulation preserves two ambitions. The functional ambition is to explain how an acquired sensory relation becomes usable and consciously accessible. The constitutive ambition is that an appropriately organized realization of the continuing self-world learning process is the experiencing system itself. The first supplies tractable experiments; its success would not automatically establish the second.

"Learning to feel" does not imply that every sensation must be learned from nothing, that learning is sufficient for consciousness, or that sensations grow monotonically stronger with practice. The particular hypothesis concerns acquired access to a specified sensory relation under declared conditions. A familiar signal could become easier to use while becoming less intrusive, or cease to require detailed conscious consideration. The proposed mechanism therefore concerns differentiated influence, not the universal maximization of activity.

2. What changes in the proposed mechanism?

2.1 Distinct meanings of significance, magnitude, and influence

The argument requires several quantities to remain separate.

Quantity	Operational question
Physical stimulus magnitude	What force, displacement, temperature, or other calibrated input was delivered?
Learned significance	What identity, predicted consequence, or possible action does the feature support?
Affective valence	Is its consequence experienced or behaviorally treated as appetitive or aversive?
Neural response magnitude	How much firing, voltage change, or spectral power was measured?
Receiver-relative influence	What changes downstream when the identified source is changed?
Experienced intensity and access	How strong is the sensation reported to be, and is its content available under the declared awareness measure?

A meaningful touch need not be emotionally positive or negative. Emotional value can influence learning and recruitment without exhausting the concept of significance. Likewise, depolarization is not pleasure and hyperpolarization is not aversion. These are different descriptions requiring a mechanistic connection.

The historical phrase *electromagnetic valence* belongs to the development of this SAN argument, including the archival note *Phase Shifts as Electromagnetic Wave Valence* [38]. Here the operational term is **receiver-relative influence**. My *Coherence-Weighted Electrical Drive in Choice and Motor Realization* supplies the earlier signed-drive framework [36]; *Transduced Valence* distinguishes neural influence from its bodily and energetic consequences [37]. These are theoretical foundations, not independent confirmations. The present contribution concerns acquisition and sensory access, not a rediscovery of those frameworks. The archival sources preserve earlier formulations without backdating the present mathematics or equating internal creation dates with public publication.

2.2 A recurrent mechanism rather than a mandatory serial pipeline

The proposed dependencies are:

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learned sensory-action relations
      -> selective recruitment
      -> receiver-relative influence
      -> recurrent sensory reconstruction
      -> inference and action
      -> returned evidence and further learning
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Value, bodily condition, expectation, inhibition, and competing activity can affect several steps. The arrows describe dependencies to investigate, not a claim that the brain executes isolated stages in that order.

"Recruitment" means that additional learned relations become consequential to the present processing. It does not necessarily mean a larger number of firing neurons. Suppressing a competing source, changing a receiving window, or selectively sustaining a relevant relation could increase useful influence without increasing total activity.

A phase-wave differential, in the phase-specific version tested here, must be defined relative to a declared ongoing reference and receiver. A difference in any recorded signal is not automatically a PWD. The relevant proposal is that the departure changes what a receiving organization does next, in the context of its history and current state.

The network's changing organization can be physically distributed in three-dimensional tissue while its relevant state has many dimensions. The analogy with artificial-network embeddings concerns relational use: a contribution depends on what it can activate or constrain. It does not equate semantic distance with anatomical distance, or show that an artificial embedding is conscious.

2.3 What learning leaves behind, and what current activity expresses

Temporary coordination and retained learning have different explanatory jobs. An oscillatory relationship can organize the current encounter. Something must also remain, or be reproducibly reinstated, for a later encounter to differ because of earlier learning. The proposal therefore distinguishes **retained receiving organization** from **its present expression**.

This is not an anatomical division between inert storage and active processing. Retention can involve interacting synaptic, cellular, and circuit processes, and timescales can overlap. The narrower point is causal: a present phase relation is not, simply by existing, an explanation of why that relation or its consequence was acquired.

My a0183z analogy argument illustrates the receiver-history problem [38]. The same explanation can connect learned relations for one listener while failing to connect them for another. The words need not carry a different physical message; the receiving histories differ. This is a theoretical example, not a measurement showing that an analogy's meaning is encoded by a particular phase angle.

Applied to touch, the retained organization determines which expectations, associations, and actions a later contact can recruit. Current coordination determines which of those possibilities become consequential now. Learned competence and momentary accessibility may therefore dissociate. The empirical challenge is to trace both, not merely to observe that activity is stronger after training.

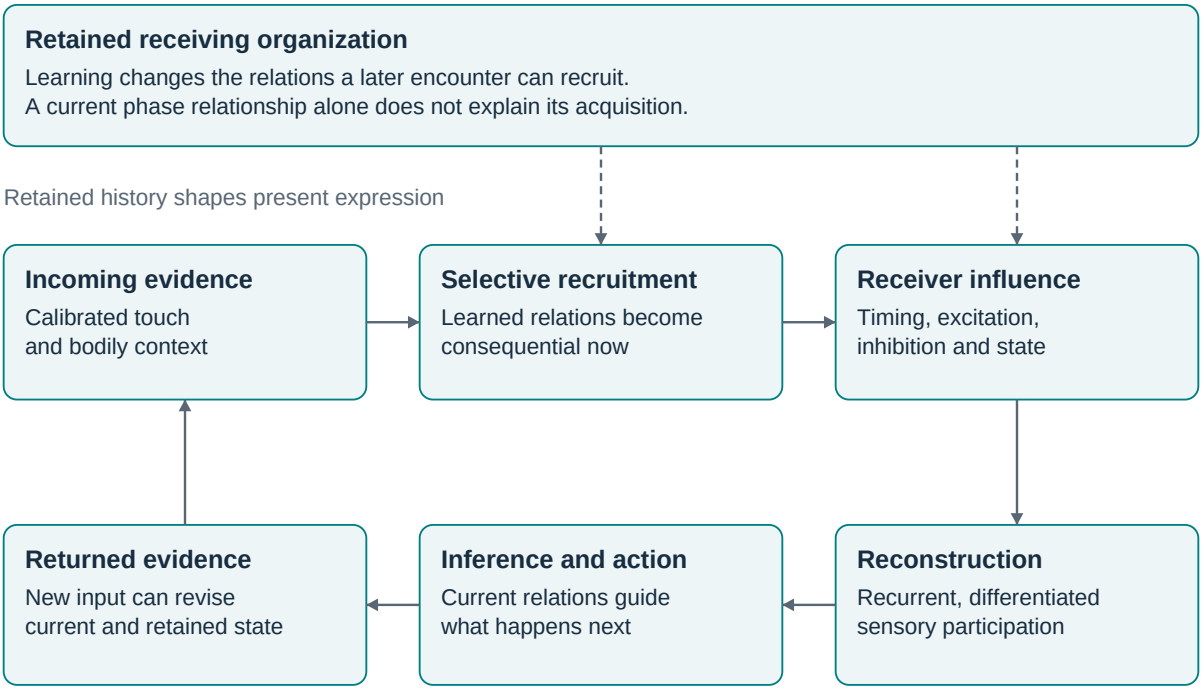


Figure 1: Retained learning and present expression. This SAN schematic separates acquired receiving organization from its momentary use. Arrows indicate proposed dependencies, not a demonstrated biological sequence; omitted reciprocal routes may operate at several steps. The six-step text loop in Section 2.2 is preserved.

3. An evidence scaffold with identifiable connections

3.1 Learning changes sensory use

Ostry and colleagues found learning-related shifts in sensed limb position without a detected change in perceptual acuity [1]. Across separate experiments they assessed position judgments during movement and with the limb stationary, observed persistence after washout, and obtained a next-day effect in an additional cohort. Their active/passive contrast was tested directly, and they examined lateral force during the probes. These converging observations support learning-related sensory recalibration. The passive comparison replayed mean movement trajectories while interaction forces differed, limiting isolation of motor agency. Dadarlat and colleagues showed that monkeys could learn to use artificial sensory feedback and integrate it with vision during reaching [2]. These results support studying acquired sensory usefulness. Neither directly demonstrates the emergence of a new subjective sense or the phase mechanism proposed here.

The artificial-feedback study makes detection and learned use separable. A preliminary task tested stimulation detectability; subsequent training made an unfamiliar spatial code usable with and without the visual movement cue [2]. Its positive contribution is learned interpretation of a changing relation that guides continuing corrections, not merely recognition that stimulation occurred. Because training combined correlated cues, reward and behavioral shaping, this result does not isolate action-contingent acquisition from other learning routes. For the present proposal, it motivates measuring the relation a signal makes available and how the organism uses it, alongside whether the signal is detected.

Recanzone and colleagues provide a tactile-training precedent: cortical area 3a responses differed between trained hands and untrained or passively stimulated controls [27]. The result supports a change in receiving organization, not the birth of a new conscious sensation. Its receptive-field changes also caution against equating learned recruitment with uniformly finer spatial resolution.

The tactile example therefore needs a controlled probe. If training changes exploratory movement, it can change the physical evidence reaching the receptors. Equal objects do not imply equal sensory input. A matched probe after training is necessary to ask what the changed receiving system contributes.

3.2 Significance is expressed through selective neural interaction

Intracellular bat recordings connect social-call categories with postsynaptic integration and selective output, without a universal mapping from affective sign to voltage sign [3]. Selective photostimulation of taste-responsive mouse amygdala populations changed licking. The available author manuscript separately describes mutual suppression in selected antagonistic subgroups, mapped without tastants after the behavioral sessions [4]. These findings support causal population effects and an inhibitory-circuit candidate, but do not by themselves establish that the mapped suppression mediated the behavioral change. They give the proposed value-to-recruitment relationship biological substance, while leaving the transfer to learned human touch unestablished.

Prior social experience was associated with later acetylcholine responses in specified mouse vocal-playback groups [49]; the dopamine experience comparisons were not significant. This links receiving history and chemical state, but does not distinguish learned vocal meaning from generalized experience or establish chemical mediation.

The proposed bridge is therefore a change in the conditions under which inputs become effective, not a conversion of emotional value into voltage polarity. Fast electrical selectivity and slower modulatory state should be linked by a receiver model with declared timescales. A measured change in one does not identify the other. For SAN, the question is whether an acquired relation changes later recruitment through those

conditions, and whether the resulting participation contributes to the independently measured sensory-access endpoint.

Popescu and colleagues add a more joined precedent: a joint amygdala-striatal gamma measure became selective for the rewarded tone and followed contingency reversal [5]. The inspected author manuscript describes that learning measure as the product of gamma powers in one-second windows. As specified, it measures joint gamma engagement and does not identify normalized spectral coherence or a learned phase change. Separate phase and spike analyses support temporal coordination, and regional inactivation supports an amygdala contribution to striatal gamma in a separate assay. These findings connect learned significance with a temporally organized network while leaving the phase-dependent contribution to learned behavior to be tested. The constant-magnitude examples in Appendix H explain why power products cannot substitute for relative-phase measurements.

Bauer and colleagues relate conditioning to amygdalo-rhinal coordination but retain learning-stage-specific effects their gamma account did not explain [6].

Headley and Weinberger connect gamma during initial conditioning with next-day responding and frequency-specific receptive-field plasticity [7]. The plasticity relationship used gamma relative to each animal's recording sites and was nonmonotonic. Initial-session associations were detected while later ones were not; that pattern motivates a direct stage comparison without itself establishing the interaction. Gamma evoked by a newly introduced unrewarded tone also predicted later bradycardia, while a corresponding relationship with discrimination between rewarded and unrewarded tones was not established. Accuracy therefore needs assessment separately from the strength of an acquired response. For SAN, the positive connection is the possibility that learned significance changes which sensory relations become consequential; the accuracy and conscious availability of those relations remain distinct questions.

Human tactile studies already join several parts of this proposed loop. Pleger and colleagues found that monetary reward improved tactile judgments and that visual outcome feedback reactivated somatosensory cortex corresponding to the judged hand, without concurrent touch [40]. Returned consequences can therefore participate in sensory processing, rather than value being confined to a separate decision stage.

An ensuing randomized, double-blind experiment in 30 participants altered reward-related effects using systemic dopaminergic manipulation [41]. This adds causal leverage, but not a localized phase mechanism or a test of acquired conscious sensation. Received reward still depended on correct performance within constant-stake miniblocks; subsequent-trial associations therefore do not separately randomize feedback.

Regional reactivation does not imply stimulus-specific feedback. In an instrumental visual task, FitzGerald and colleagues' fMRI adaptation analysis supported a nonspecific outcome component in stimulus-responsive cortex, without excluding concurrent stimulus-specific feedback [42]. This is an indirect population-level inference, not a recording of synaptic tags. Shindou and colleagues supply a cellular precedent in mouse striatal slices [43]. Delayed dopamine relieved depression without nimodipine; net potentiation was reported when nimodipine blocked depression. The reported potentiation statistic contains an unreconciled t/P inconsistency, so I do not use its numerical significance or magnitude. Its rectification marker also outlasted the tested effective plasticity window. This is a qualified precedent, not a calibrated trace duration, tactile-awareness result or oscillatory implementation.

A closer sensory bridge comes from Hong and colleagues' in vivo mouse V1 experiments [44]. Neuromodulator pairing produced stimulus-specific response changes, and disrupting receptor anchoring interfered with conditioning. Norepinephrine-associated orientation changes occurred in awake mice; the corresponding serotonin depression was observed under anesthesia after awake tests failed to show it. Off-target effects remained possible. These results connect timed neuromodulation, receiving machinery and

later sensory response, rather than merely placing those topics in adjacent literatures. They do not establish human touch, phase-mediated recruitment or conscious access.

Supplementary controls refine this bridge [44]. Conditioning the alternative orientation and comparing the two eyes supply tests of input specificity, not proof that every unpaired response is unchanged. The serotonergic intrinsic-excitability control used posterior piriform cortex because V1 did not show the agonist response in that assay. It therefore does not provide a same-preparation V1 exclusion of that alternative.

Jordan and Keller supply an action-contingent bridge [53]. In mice, periodic locus-coeruleus (LC) axon stimulation accompanied either coupled movement and visual flow or an open-loop replay. Subsequent fixed-speed probes revealed a direction- and locomotion-dependent change in V1: responses to flow associated with forward movement were reduced during locomotion, with stationary responses comparatively stable. The corresponding pattern was not detected after the open-loop condition or in low-expression and noninjected controls. This connects acquisition conditions to later sensory expression within one preparation, without requiring emotionally valued stimuli.

A separate assay in the same study found only small immediate effects on average measured responses. These assays support distinguishing plasticity from a large immediate population-wide amplification. Selective suppression of predictable self-generated input could change what other evidence becomes consequential; that further downstream benefit was not demonstrated here. The study does not establish long-term retention, a uniquely local norepinephrine mechanism, phase mediation or conscious access. For SAN, its positive contribution is a concrete acquisition-to-reception join, with the subsequent use and experience claims still to test.

The statistical supplement reports direct comparisons of learning-related locomotion-modulation changes across conditions [53], strengthening this inference beyond a comparison of separate significance labels. The somatic analysis samples imaging sites and neurons within a small number of animals; its animal-level dependence remains unresolved. I use the qualitative operation, not its reported P values as a calibrated effect or planning estimate.

For SAN, the connecting claim is that outcome-related re-entry can update a cue's retained receiving organization and thereby change its later influence under matched stimulation. That is stronger than identifying a current reward response. I distinguish feedback during acquisition, reactivation during outcome evaluation, retained change, and expression during a later unrewarded encounter. The proposed experiment must trace the transitions, rather than read durable learning or consciousness directly from reactivation.

The receiving-state link can therefore involve a broad teaching signal acting on connections made selectively modifiable by their recent history. This transient local eligibility is distinct from durable learning. A further question asks how the changed relation is later recruited under a current phase/reference. Assigning credit during learning and expressing learned content through a receiving window are different operations and may use different timescales. In the proposed SAN synthesis, these operations connect the returned consequence to what a subsequent encounter can recruit; they should not be collapsed into one undifferentiated increase in activity.

Learning, significance, oscillatory coordination and feedback into sensory processing already have joined experimental precedents. The added in vivo bridge further narrows the contribution: this paper does not newly propose that neuromodulation can selectively reshape sensory responses. I propose how acquired receiving relations become available to a later, competing, recurrent rendering and sensory access. The transition from retained change to its present use is the connection to test, not an established conclusion borrowed from a plasticity experiment.

3.3 Timing is more specific than synchronization strength

Kanta and colleagues used phase-triggered stimulation during post-learning consolidation to alter later spatial-memory performance. Both phase treatments increased spike locking, but with different preferred phases and consequences [8]. This is a causal timing precedent, not evidence for online tactile awareness. Their random-timing control did not retain the burst gate used for phase-targeted stimulation. Thus, random timing and phase specificity must not be treated as interchangeable control definitions.

Siegle and colleagues provide a sensory counterpart: gamma-range manipulation could improve weak tactile detection under particular timing conditions, without a universal benefit across stimulus strengths [9]. Their discussion explicitly leaves open whether an isolated synchronization event can provide the useful receiving window without subsequent cycles. The lesson is not that gamma has one psychological meaning, or that repeated oscillation is thereby necessary. It is that a signal's consequence can depend on its relation to a receiving process.

In their recorded layer II/III sample, entrainment reduced mean firing and increased individual-unit vector strength across timing conditions [9]. The beneficial alignment also had a more concentrated early multiunit response. Increased locking alone therefore did not identify the beneficial alignment. Moreover, the measured rate and timing changes did not explain the behavioral impairment at another offset; sparse-cell gain and downstream transformations remained possible contributors. For SAN, this supports testing a pattern's effective influence beyond its sampled mean activity or rhythmicity. The question is which organized contributions become consequential to a receiver. Because stimulation began after task training, the acquisition of that receiving relationship remains a further link to investigate.

In four cat and monkey recording datasets, Womelsdorf and colleagues conditioned between-site oscillatory-power correlations on relative phase under unchanged visual/task conditions [28]. Their timing, frequency and recording-triplet analyses support structured, pair-specific coordination beyond a coherence value alone. The favorable phase was defined for the recorded pair, not as a universal zero-phase target. This gives the proposed receiving relationship a concrete observational precedent; the statistic is not itself an intervention-defined input-output effect. Selective perturbation and an independently measured downstream consequence remain necessary for the causal claim tested here.

A relevant contrary comparator is Histed and Maunsell's task in which trained mice detected direct excitation of V1 populations [29]. Across a 100-ms stimulation window, behavioral thresholds followed integrated light input despite changes in pulse concentration and trains. Their recorded population response was approximately linear. Most quantitative physiology was collected awake but outside the detection task, after a comparison of those recording contexts. The experiment varied excitatory drive; the tactile study instead aligned peripheral input with a manipulated inhibitory rhythm. These differences define a comparison to investigate, rather than establishing why the results differ. SAN must specify the receiving conditions under which a temporal relationship changes downstream use, while allowing integration to account for successful detection in other conditions.

For the proposed experiment, measure mean firing, local power, individual-unit locking, interregional phase relationships, early population-response width and downstream influence separately. Declare their references, averaging intervals and sampled populations. A unit's periodicity, a between-site coherence estimate and an intervention-defined input-output effect describe different relationships. The proposed learned receiving mechanism should explain beneficial and adverse timing conditions, including cases where integrated input is sufficient. This makes the connection from retained learning to its current use an experimentally discriminating claim.

3.4 Recurrent reception and conscious access

Manipulating an S1-M2-S1 feedback circuit affected tactile processing and performance in mice [10]. Manipulating apical dendritic activity shifted whisker-detection thresholds [11]. These are candidate receiving and feedback operations. They do not identify the complete learned, value-sensitive pathway proposed here.

In the feedback study, behavioral disruption followed training; acquisition itself was not manipulated [10]. Physiology and behavioral assays used different preparations and intervention windows. Their convergence supports investigating recurrent reception, but does not supply one trial-level chain from learning to access. A learned receiving relation and its acute contribution must therefore be tested separately before their connection is inferred.

The dendritic study strengthens this receiving-to-behavior join [11]. Suppression raised detection thresholds, whereas direct excitation lowered thresholds and increased false alarms. The latter does not uniquely distinguish changed perceptual evidence from decisional or motor contributions. The comparison of trained and naive dendrites also involved different groups and task states, not an isolated acquisition manipulation. Thus, consequential apical reception is an experimentally grounded component; its acquisition through the proposed learning loop remains a connecting hypothesis. Apical amplification is already part of that study's explanation, not a mechanism first introduced here.

Human studies link reported access to distributed neural coordination [12,13], while other experiments dissociate particular gamma/P3 measures from awareness when task demands are separated [14]. These findings are not interchangeable observations of a single "consciousness signal." They motivate a study in which awareness, task relevance, memory, and reporting are varied or measured independently.

The measurements also differ. Melloni and colleagues assessed between-electrode phase-difference consistency in masked-word matching, not a phase intervention or a separate subjective rating on every trial [12]. Pitts and colleagues' phase analysis instead assessed individual-electrode consistency across stimulus-aligned trials [14]. Their task-irrelevant shape-minus-random gamma contrast was significant even though the shape response did not differ significantly from baseline. Thus, absence relative to baseline and absence of a stimulus contrast are not equivalent. These findings motivate distinct neural and access endpoints rather than one scalar consciousness signal.

The sampled episode also changed over time. Melloni and colleagues found a brief early between-electrode phase effect; P3a-like, retention-related theta, and test-expectancy effects supplied further distinctions between the visibility conditions [12]. The long window needed for theta prevented ordering its onset precisely against the gamma episode. The authors left open whether phenomenal awareness corresponded to the early coordination or the wider ensuing process. Within SAN, I propose a candidate succession in which transient recruitment conditions later receiving and reconstruction. In this hypothesis, related successor states preserve and transform the encounter as the participating network changes. Testing that connection requires identifying what the coordinated state changes next.

Three further visual studies sharpen the access assay. A two-interval detection/discrimination comparison favored an excitability-related shift in subjective evidence over the specified strategic-decision explanation [33]. A contrast-comparison study related lateralized alpha to perceived contrast [34]. A later reanalysis found network-specific alpha relationships with sensitivity and criterion [35]. These associations do not identify a learned tactile phase mechanism. They do show why subjective magnitude, sensitivity and decision strategy must remain distinct: a signal-detection criterion shift is not automatically a report-only artifact, and alpha need not have one function across all receiving networks.

An animal's learned response establishes an operational achievement in that preparation. It should not be

silently substituted for a human phenomenological report. Conversely, a verbal report alone does not identify the neural mechanism producing access.

3.5 Awareness during acquisition is not the same as later expression

Seitz and colleagues demonstrated reward-associated perceptual learning under conditions designed to suppress the training stimuli [15]. Their awareness checks were conducted after training and in a separate naive control group, rather than continuously establishing each learner's awareness on every acquisition trial. This supports a bounded learning-without-reported-awareness interpretation; improved later orientation discrimination is not itself a direct measure of newly conscious sensation.

Dong and colleagues investigated reward associated with the eye of origin, a delivery attribute distinct from seeing the target [16]. Their eye-specific breakthrough advantage during training was not detected in the unrewarded post-test. This is an important learned-access precedent, but not evidence of durable, context-independent enhancement. Neither study measured the SAN phase mechanism.

The hypothesis does not require conscious supervision of every learning event. Nor does it require a self-generated action for every acquired sensory relationship: task-free reward pairing remains a serious alternative acquisition route [15]. Whether executing an action adds to informative exposure is an experimental question. The present proposal must separately identify what was learned, what was consciously available, and when the learned relationship affected performance. A null awareness check does not establish complete unawareness, and disappearance of a response advantage does not uniquely identify erasure of retained learning.

An attention/adaptation experiment further distinguishes present selection from its later consequences. After prolonged viewing of competing movies with attention directed to one eye, Song and colleagues observed an ocular-dominance aftereffect favoring the previously unattended eye [48]. Preadaptation continuous theta-burst stimulation over the frontal eye field reduced this aftereffect relative to a vertex control. Concurrent blob detection showed no detected stimulation-site effect; this coarse task does not establish unchanged attention. The direct frontal-eye-field versus intraparietal-site comparison was not significant. For the present hypothesis, this motivates separating current selection from retained change, not treating every aftereffect as amplification of the attended input. It does not establish durable reward learning or the proposed phase-dependent route to conscious tactile access.

Learning-stage interventions add a complementary bridge. In Pavlovian whisker-reward learning, Moberg and colleagues found comparatively stable intratelencephalic (IT) population responses and developing extratelencephalic (ET) reward-expectation responses [54]. Silencing either subtype during training impaired discrimination, with reduced anticipatory licking after IT suppression and generalized responding after ET suppression. S1 silencing produced no detected impairment in experts; this is not an equivalence result. Reward was independent of licking, the stimuli were suprathreshold, and trunk calcium indexed global cellular activity rather than selectively measuring local dendritic spikes. The study's stage-specific effects do not concern oscillatory phase. Its contribution here is that a route's role in establishing a learned relation can differ from its role during established performance. Neither that result nor an acquired anticipatory response establishes conscious sensory access.

3.6 Persistent changes constrain the learning bridge

In mouse motor-learning studies, training-related dendritic spines could be selectively stabilized beyond the training period [25]; complementary work linked experience-dependent remodeling and stable cortical connections [26]. These are concrete precedents for retained changes, not evidence that every learned meaning is stored in one spine or that stable anatomy by itself produces access.

For the proposed scaffold, the additional inference is that retained learning changes subsequent receiving and recruitment, which current coordination expresses. Neither spine persistence nor gamma modulation alone demonstrates that complete link. The experiment needs delayed probes and dissociations between retention and expression.

3.7 What the connected evidence supports

The constructive argument is that learned significance need not be an additional substance placed on an electrical signal. It can be partly realized by the relations that a signal becomes able to recruit, the consequences those relations predict, and the actions they make possible. Retained learning changes this repertoire; current receiving conditions determine its expression. In SAN, recurrent exchange then permits the recruited relations to constrain the developing sensory situation, rather than remaining information available only to an outside decoder.

The evidence supports this explanation by constraining different parts of it. Learning can change sensory use and cortical organization. Outcome feedback can also re-enter task-relevant human sensory processing [40,41]. Value-related populations can change behavior. Relative timing can alter reception, while recurrent pathways can affect sensory performance. Coordination also has experimentally observed relationships with reported access. These findings make the proposed joins biologically grounded questions rather than unconstrained metaphors.

The inference across studies is functional, not an assertion that bat amygdala, monkey somatosensory cortex, and human access tasks implement an identical circuit. The transferable variables are learning history, cue-dependent recruitment, receiving state, downstream consequence, and the measured outcome. Their anatomical implementations and timescales must be estimated for each preparation. A valence manipulation is relevant when it changes a cue's consequence; it need not imply that every meaningful cue has emotional valence.

The new causal claim is that these operations connect within a specified learned sensory relation. Section 6 tests that claim jointly. The further claim that the appropriately organized process constitutes experience remains explicit in Section 8; it is not obtained merely by treating access reports, firing, and subjective experience as interchangeable endpoints.

4. Relationship to existing theories

Communication-through-coherence already proposes that rhythmic receiving conditions organize effective transmission. Fries's later formulation develops directed entrainment, inhibition, and competition among converging inputs [17,18]. A strong comparison must include these operations. The proposed SAN mechanism cannot earn novelty merely by replacing their names with PWD terminology.

Sensorimotor theory already places mastery of action-dependent sensory relations at the center of experience [19]. The overlap with perception-action learning is substantial. The theoretical disagreement concerns whether a recurrent internal construction is constitutive of experience. Describing that disagreement does not yet distinguish the accounts experimentally.

Recurrent-processing accounts distinguish feedback-dependent processing from an initial feedforward sweep [20]. The global-workspace model already connects distributed selection, reward-related learning, and descending modulation without a literal central observer [21]. Consequently, neither recurrence nor distributed access alone is a distinctive contribution here.

Active inference supplies a formal account relating action, preferred outcomes, and information-seeking [22]. Its precision variables should not be treated as synonymous with physical voltage or power. A mathematical comparison requires the particular model's equations, rather than a contrast between loosely described schools.

Whyte and Smith's predictive global neuronal workspace explicitly models visual access through hierarchical active inference [45]. A later account includes a reward-biased rivalry simulation with a precision trade-off [46]. These are joined formal alternatives, not just general references to prediction. Their simulated access depends on declared links between model variables and report; it does not establish experience in the simulated system.

The published minimal-theory account [46] additionally proposes that a suitably precise representation becomes conscious through its capacity to inform counterfactual policy selection at an appropriate temporal scale, located at the continuous-discrete interface. It restricts its sufficiency claim to humans and similar neuronal architectures. This is an explicit experience hypothesis with proposed neural mappings, not an experimental observation of consciousness in an arbitrary controller. My learned receiving relation can supply a physical implementation constraint to compare within or alongside that account: identify what learning changes, how an input becomes consequential now, and whether disrupting that relation changes the separately assayed access outcome.

The Motivated Emotional Mind preprint also joins associative memory, selection, action, value and sensory re-entry, and proposes tests separating reconstructed quality from report and global availability [47]. This is substantial architectural overlap, not independent experimental confirmation of SAN. My proposal must therefore specify the acquired receiver relation and its consequences, rather than claim that combining learning, embodiment and reconstruction is unprecedented. A shared mechanism could contribute to both accounts; different terminology alone would not separate them.

Reward-modulated learning with activity-dependent eligibility already explains how a broad teaching signal can yield selective change [42,43]. That operation is a foundation, not this paper's novelty. The proposed addition is its connection to later consequential recruitment and sensory access; an eligibility trace alone does not establish that connection or distinguish SAN from an account sharing the same learning rule.

The learning-stage study also supplies a concrete conventional comparator [54]. Its Rescorla-Wagner-based model uses stable sensory representations, predicted value and prediction-error updating; a separately trained non-S1 route supports expert performance without S1. The authors acknowledge that this explanatory model is not unique, and the biological prediction-error circuit was not identified. A SAN comparison must allow such teaching-versus-expression roles and alternative learned routes. Reproducing acquisition and expert behavior alone cannot establish the additional phase-dependent receiving and access claims.

The residual sought is specific: **an acquired sensory contingency changes receiver-relative recruitment and timing, and an identified part of that change contributes to access under controls that distinguish gain, generic recurrence, task performance, and report.** Other accounts may predict this too. The study must compare those predictions directly, and a successful result would support the tested mechanism before establishing a theory-wide advantage.

A later adversarial collaboration tested preregistered biological predictions of global-workspace and integrated-information theories [39]. Its phase-consistency test supported neither theory; complementary connectivity analyses yielded a different pattern. This is not a SAN test or a declaration that either entire theory failed. It makes a methodological requirement concrete: define the metric and expected result before testing, and distinguish later explanatory analyses from confirmation of the original prediction.

4.1 What would discriminate the proposed connection?

Competing explanation	What it already explains	Additional test needed here
Exposure or informative associative learning	Acquisition without executing the original action	Contingent training versus replay retaining action/context information
Ordinary gain or mixed-population weighting	Stronger responses or an apparent aggregate tuning change	Receiver-level phase functions, declared response scale and reference, heterogeneous-gain comparison
Changed phase-delivery fidelity	A larger contrast between assigned timing policies without stronger intrinsic receiving modulation	Declare the reference and achieved-phase uncertainty; separate the randomized policy effect from calibrated receiving modulation
Generic temporal memory or recurrent inference	Sequence-sensitive decisions and reconstruction	Matched nonoscillatory and direct-inference alternatives, plus physical interventions
Altered subjective evidence or decision strategy	Changes in reported detection	Joint hits/false alarms, magnitude comparisons and complementary tasks
Task-dependent distributed availability	A relation becomes accessible for reporting and action	Prespecified receiver contribution under independent task/report controls
Preference- and precision-dependent inference [45,46]	Value, expectation and sampling can change modelled access	Joint held-out receiver and access predictions under declared intervention mappings, not a reward-blind baseline
Learned embodied reconstruction [47]	Retained associations can support sensory re-entry, value and action	Specify the acquired receiving route and compare its necessity with alternative realizations of the same function

The target is not to dismiss these explanations. Several may implement parts of the proposed SAN process. The discriminating claim concerns a specified learned route and its incremental explanatory or causal value. No current prediction is claimed uniquely diagnostic of SAN, and shared evidence cannot decide between indistinguishable implementations.

4.2 Compare joint predictions under shared observation rules

A next model comparison should use the same acquisition histories, physical probes, reward/context information and reporting demands. Freeze each model's learning rule, retained state, observation equations and intervention mapping before confirmatory outcomes. An inference model must be allowed the learning, expectation and temporal information its stated formulation supports; a recurrent comparator must not be denied feedback merely to favor SAN. Match usable information and report capacity differences rather than claiming parameter equality where none exists.

The added constraint sought here is that an independently measured receiving relation predicts both the downstream response and the learned access effect of disruption or restoration. Estimate parameters from training and designated calibration data, then evaluate held-out interventions and contexts with prespecified predictive scores and uncertainty. Fit complementary behavioral outcomes jointly; allow decision-policy alternatives rather than interpreting every report change as altered sensation. Do not turn a fitted likelihood

precision into a measured phase angle or identify an observational mediator merely because two predictions improve together.

The qualified source and author implementations [45,46] make the intervention mapping concrete. Distinguish a change in normalized likelihood, parameter concentration, transition belief, retained state, current inference, report preference and recording transform. A receiving intervention must alter the nominated causal operation before its predicted consequence is scored; changing only a completed neural-proxy waveform addresses observation. Freeze the operative equations, settings, dependencies and measurement reference. Appendix J shows why matching likelihood means or reconstruction means does not guarantee matching learning, uncertainty or information seeking. Exact source settings and unresolved article/code differences belong in that qualification, not in an assumed biological calibration.

If models predict the same joint observations, report the unresolved equivalence. If a comparator predicts the intervention effects without the proposed phase-specific route, that result weakens its claimed necessity while potentially supporting learned recruitment more broadly. If the constrained receiving model predicts the effects better, the result favors that implementation under its assumptions; it does not establish the whole SAN architecture or settle phenomenal experience. This is a prospective analysis requirement, not a completed comparison of these external theories or another acquisition arm.

An executed source-model qualification now makes part of that comparison operational [55]. The hierarchical report/no-report model retains task-dependent inference and memory control; its implementation is not a reward-blind alternative. Fresh probes receive declared learned parameters rather than cached observations or transformed task quantities. A separate adaptation distinguishes received correctness evidence, anticipated information availability and the utility of correct responding. Section 7.5 and Appendix K declare its exact controls and limits. This qualifies a computational interface for later comparison; it does not identify a physical receiving relation or reproduce every published simulation.

Section 7.4 implements a limited synthetic version of this measurement comparison: receiver change, policy change and partial compensation share an explicit observation model. It does not reproduce the full external theories above. Its additional measurement separates some stage explanations while leaving exact implementation equivalence and causal mediation unresolved.

5. Formal consequences of the proposed mechanism

The technical supplement retains all 21 equations, eleven conditional propositions and their proofs. Four distinctions guide the main argument.

First, **influence needs an endpoint**. The response of a declared receiver to a controlled change in a source is different from correlation or coherence. The candidate causal quantity has units determined by those variables and a specified lag. An observational association alone does not identify it.

Second, **a learning-by-phase effect need not be retuning**. A first-harmonic response has an intercept and a two-dimensional phase vector. Uniform positive gain changes its magnitude without changing its direction. A nonzero determinant between pre/post vectors rejects scalar gain under the stated model, but reference drift, response-link errors or mixing heterogeneous receivers can produce apparent departures. A zero determinant also permits reversal or an undefined zero-baseline direction. These are conditional identification results, not a claim that every neural response is sinusoidal.

Third, **retained learning and present expression have separate states**. A current state evolves through input, action and returned evidence under retained parameters; learning changes those parameters. The implemented temporal receiver makes this distinction explicit, but its finite-window phase map has an exactly

equivalent time-bin readout. Informative replay also gives identical weights under the same deterministic update. The mathematical results therefore constrain what successful behavior can identify.

Fourth, **sensory reports do not uniquely identify their cause**. In an equal-variance Gaussian observation model, hits and false alarms identify sensitivity and an effective criterion. They do not separately identify a common shift in internal sensory evidence and a change of decision threshold. Unchanged sensitivity therefore need not mean unchanged felt magnitude. Complementary measures are required, while their own assumptions remain visible.

The reconstruction example supplies a further conditional result: learned statistical relations permit partial evidence to change estimates of missing content, and a new independent observation reduces model uncertainty. Repeating the same observation as if independent does not supply that evidence. Its recurrent coordinate solver converges to the same solution as direct inference, so the example demonstrates a function without identifying a uniquely recurrent physical implementation.

A fifth application adds three identification results, developed in Appendix E. Two explanations may predict identical behavior while differing in a calibrated receiver observation. Conversely, phase features and stored temporal coefficients can yield identical joint predictions. Even a correct joint distribution may admit either mediation through the measured receiver or a bypass driven by the same upstream state. A selective intervention with justified exclusion assumptions is therefore distinct from successful joint prediction.

These are inspectable mathematical constraints on the theory and its tests. None derives phenomenal experience or establishes the proposed biological pathway.

A post-result sensitivity analysis in Appendix F asks whether the observation mapping is stable. If a recording change follows the tested perturbation, the measurement can identify a sum of biological and recording effects rather than either alone. A justified calibration bound can still constrain the biological contribution. The derivation separates this ambiguity from fixed changes of measurement units and from causal mediation; it does not assign a biological nuisance bound.

A complementary phase-delivery analysis asks which physiological phase was actually reached. Under a first-harmonic conditional-mean model and a target-independent phase-error law, the observed harmonic amplitude is the product of intrinsic receiving modulation and the magnitude of the phase-error distribution's first circular moment. That error concentration is not inter-neuronal coherence. Improved targeting and increased receiving modulation can therefore give identical assigned-phase mean curves. They need not give identical full observation distributions. Applicable delivery bounds can constrain the intrinsic component; an assumed or unstable bound cannot. These elementary results add no fitted application, physiological estimate or proof of experience.

The allocation and delivery analysis adds three elementary design results. Yoke-set dependence and generator-response congruence distinguish the role-policy effect from a direct benefit of action execution at fixed exposure. Presentation availability can produce both learning-by-timing contrasts without changing responses to a delivered event. For incomplete binary observations, finite-cell completion bounds show what retaining assignments can and cannot establish. These refinements add no fitted application or biological effect, and do not replace the preceding identification results.

The phase-relation and access analysis further distinguishes alignment to a stimulus from a relation between receiving populations. Identical individual phase distributions and magnitudes can accompany different between-site consistency. Conversely, perfect relative consistency can arise through either a directed route or a common drive, with different responses to an ideal selective intervention. Finally, a block-level awareness answer and neural mean do not identify trial-wise awareness-conditioned responses. These elementary constructions refine what the proposed measurements can establish; they add no learned application or biological result.

A further inference-mapping comparison makes the shared operation explicit. Under fixed linear-Gaussian assumptions, variational inference and Application 04's quadratic reconstruction have the same optimal mean; restricting covariance can change uncertainty while preserving that mean. Softmax logits retain a common-offset freedom, so an uncalibrated latent sign cannot be imported as membrane polarity or affective sign. Equal normalized likelihoods also permit different parameter-learning and novelty weights. These elementary results preserve the constructive examples and locate the extra responsibility of a physical learned-reception and access test. They do not add a fitted application or establish equivalence of the complete theories.

Appendix K adds three elementary conditions for the executed comparator interface [55]. A zero scalar learning rate need not freeze every count register; an unreceived channel cannot generally be equated with a recorded null outcome; and a task utility can remain when a correctness observation is withheld. The controller must define which returned changes persist. These are specified model and observation properties, not new biological learning laws or derivations of experience.

6. Test acquisition, influence and access together

6.1 Acquisition and matched probes

Randomize three acquisition policies: active contingent training, informative yoked replay and active noncontingency. Replay preserves the generating action/context/outcome relation as well as the stimulus stream; otherwise the comparison removes information rather than isolating action execution. Compare contingent training with each control separately. Yoke-set dependencies and residual differences in agency, engagement and sensory state must remain explicit.

Use fixed calibrated probe intensities and catch trials before training, immediately afterward and after a prespecified retention interval. Include passive matched-input probes, untrained features and a planned contingency reversal. Matching an object or movement trajectory does not ensure equal force or receptor input. Do not re-titrate away the target learning effect or adjust away training-induced state changes in a primary total-effect estimate.

Within those existing probes, score detection, decoding of the learned relation and its use in the next action separately. A later cue-only failure can reflect unfamiliar task instructions, so any preparatory cueing or response training must be fixed before the comparison and recorded, not counted as a sensory benefit. Predeclare the evaluation window and retain non-initiation separately from incorrect initiated responses. This refines the observation protocol; it adds no acquisition arm and does not identify conscious experience by successful decoding.

Because the contingent participant generates the yoked stream, changing the role allocation can change other participants' exposure. The primary C-versus-R and C-versus-N estimands therefore concern the complete declared allocation and replay policy. A yoke-set hierarchy accounts for dependence but does not isolate action execution with exposure held fixed. Appendix I gives a generator-response congruence counterexample: identical fixed-stream response rules can still yield positive contrasts against both control roles. Prespecify stream generation, role allocation and prior-history measurements. A separate design using streams fixed independently of confirmatory role assignment may address an execution contrast, while retaining its receiving-state and input-matching limits; it is not a fourth arm or a replacement for contingent learning.

Keep acquisition, consolidation, receiving, report and restoration windows separate. A perturbation of consolidation may change a later trace without controlling online sensation. Retention without present expression, and restored expression without relearning, are distinguishable outcomes.

An optional stage-specific circuit companion can distinguish these causal jobs [54]. Compare acquisition-only interruption followed by independently validated recovery and matched probes with probe-only interruption after intact learning; treat acquisition and probe interventions as separate factors. Measure learning opportunity and generalized responding, so impaired task engagement is not mistaken for selective failure to establish a relation. Then compare established performance with reversal or transfer requiring new updating. Declare the proposed teaching route, present-use route and plausible alternatives before observing results. Acquisition-only and probe-only deficits support different roles; precise preservation after a validated selective interruption challenges that route's necessity at that stage. Prespecify meaningful-effect bounds before claiming preserved function. These are refinements of P1-P4 for a feasible circuit preparation, not a fourth acquisition policy, a new human invasive procedure or a requirement to cross every condition in the main study.

Prespecify the sign, feature specificity and duration of change at each probe rather than defining every aftereffect as strengthened access. If an intervention's effects span acquisition and testing, a later difference cannot alone locate the causal stage. Independently verified intervention windows and matched assay controls are needed before attributing it to changed acquisition rather than altered expression. These are interpretation requirements for the existing arms, not an additional acquisition policy.

The sensorimotor comparison further distinguishes acquisition coupling from receiving state at the later probe [53]. A feasible circuit companion should test whether the acquired feature-specific change is expressed differently across declared movement or context states, using direct contrasts rather than significance in only one state. Prespecify whether the candidate operation predicts enhancement or suppression and measure its downstream use separately. This does not add an acquisition arm or license conditioning the primary total-policy effect on a state changed by training. A later state-dependent response is evidence to explain, not an access measure by itself.

The primary immediate and delayed probes should withhold feature-contingent reward and correctness feedback, while preserving ordinary compensation and matched physical input. A separate, prespecified context comparison can test whether the learned relationship is expressed differently when its acquisition context is reinstated. Assess the initial probe window before new differential feedback; otherwise renewed reinforcement can produce reacquisition. Context order, expectation, response policy and carryover remain alternative explanations. A reward-free null is not proof of erasure, but neither may an unobserved trace be invoked to protect the hypothesis from every failure. Prespecify the feature-specific retention or recovery that would support the claim and the precise failures that would constrain it.

The computational counterpart must also distinguish wanting a correct response from receiving correctness feedback [55]. Our declared adaptation holds the task utility and saved upper-level learning parameters fixed, makes withheld terminal feedback neutral evidence, and removes announced unavailable feedback from anticipated information. Its policy effect includes preparation for that availability condition; it is not solely a late observation deletion. The source's initial-state counts can still update at zero scalar learning rate, so probe returns are discarded instead of silently replacing the acquisition snapshot. This limited qualification does not implement the full human reward-free assay or a complete hierarchical acquisition rule.

Acquisition-awareness measures must name their target: cue presence, feature identity, contingency knowledge or a hidden delivery attribute. Retrospective reports and separate cohorts do not identify every acquisition trial. Adding probes can itself alter attention or learning, so their timing and interpretation require independent pilot work.

A dedicated feedback-credit follow-up can test another link: does a returned outcome change later reception when the next input and incentive are held fixed? Randomize a disclosed bonus only after a prespecified eligible judgment, retaining accurate correctness feedback and guaranteed ordinary compensation. Compare

later reward-free, matched-input probes under the assigned bonus, with eligibility fixed before that randomization. This identifies the feedback-policy effect within that eligible population, not a total acquisition effect or a uniquely neural mediator. Keep this follow-up separate from the primary three-arm study until feasibility work establishes how its extra feedback would preserve the yoking and information controls.

The stronger mechanism predicts more than a favorable next trial. A feature-specific influence change should remain available at a prespecified later probe, or show the declared context-dependent recovery, and the receiving-window test should establish whether its expression depends on the proposed timing relation. A transient improvement without retained or recoverable specificity supports a current state effect, not that retained-change claim. Reported access remains a separate endpoint.

The feedback follow-up should distinguish cue-evoked activity, outcome activity and later cue-specific receiving change. Regional overlap among them is insufficient. Define cue contrasts and recording regions using independent calibration or held-out data, and compare the assigned-bonus effect on the credited cue with its effect on a counterbalanced credit-irrelevant cue. Test this interaction directly; significance for one cue but not another is not an interaction test. Broad feedback followed by selective retained change is compatible with local eligibility. Neither that pattern nor an adaptation-related spatial correlation uniquely identifies its cellular implementation. A feedback-delay effect alone is also ambiguous because delay can alter expectation, attention and intervening processing.

For the circuit component of this research program, measure net response change separately from proposed opposing plasticity processes. A larger response than a depressed control need not exceed the original baseline. Conversely, a near-zero net change does not identify absent versus offsetting modifications; an independent assay or adequately selective intervention must distinguish them. Do not infer a hidden trace after every null. Define the tested physiological state, verify the delivered modulator's time course rather than equating a trigger with its arrival, and test a proposed eligibility marker against functional modification at prespecified delays. The human follow-up retains its direct cue-by-assigned-bonus contrast; no additional acquisition arm or universal molecular clock is introduced.

6.2 Receiving conditions and access measurements

Choose the source, receiver, lag, reference and neural metric independently of confirmatory outcomes. Address common input, leakage, evoked responses, power/SNR, movement and phase-estimation error. A change in synchronization is not itself evidence that information was used. The metric cannot be replaced after an adverse result and the replacement called confirmation.

State whether the prespecified phase quantity averages across trials at an event-relative time, across time, or between sites. These operations need not measure the same organization. Separate task relevance from executing a response on a particular trial; a nontarget can remain relevant to classification. Where feasible, vary task and report demands independently and test interactions directly. Delayed queries can alter expectancy and memory. A block report constrains its declared aggregate endpoint, not every trial's experience; reduced reporting still requires a justified access measure.

Prespecify which operation a perturbation is intended to test: initial reception, retention, or comparison with a later input. A deficit in delayed matching after a late intervention could reflect altered retained availability without establishing that an earlier encounter was not experienced [12]. Where feasible, compare independently calibrated receiving and retention windows with endpoints appropriate to each operation. Calibration must establish those windows in the actual preparation; latencies from the masked-word EEG study are not transferred to the tactile assay.

Validate the observation mapping under the tested perturbation and learning conditions, not only in an unperturbed baseline. A perturbation-aligned recording component can imitate a receiving change while remaining absent from baseline controls. Preparation-appropriate independent calibration and negative controls should constrain that component; their uncertainty and transfer assumptions must enter interpretation. Genuine learning-related changes in recruited populations or receiving state remain biological effects of interest, not nuisance variables to regress away. If applicable calibration bounds exclude a recording-only explanation, they strengthen the evidence for receiving change; the downstream-use and mediation tests remain separate.

If an inference model supplies the neural interpretation, declare the electrical reference, units and conversion from its latent state to the recorded receiver endpoint. Relative logits, posterior uncertainty, likelihood concentration, phase-error concentration and physical response magnitude have different definitions. Calibrate their proposed links under the tested learning and intervention states. Appendix J's reference and concentration results explain why a model label alone does not supply those links. This retains the positive possibility that acquired relations are physically realized through changed reception and recurrent use while keeping their measurement inspectable.

When evaluating a proposed neuromodulatory intermediary, distinguish an input to the receiving state from a response produced by the same underlying event. Their correlation is compatible with both arrangements. In a feasible circuit preparation, perturb the candidate receiving process while verifying comparable sensory input, and test its downstream consequence separately from its association with behavior. Reassess the relevant state at matched probes; do not remove a treatment-induced state from a total-effect analysis and then call the residual the whole learning effect. These are tests of the proposed intermediary, not an additional human intervention or acquisition arm.

For a competition-mediated account, record target and opponent populations during the same behavioral condition. A separate connectivity assay does not supply that mediation result. Where feasible, selectively prevent or restore opponent suppression while preserving the targeted drive, sensory evidence and motor competence. This refines the receiving-pathway test rather than adding another acquisition arm.

Compartment calibration must distinguish direct somatic interference from the somatic consequence of changing dendritic influence [11]. Selective action on a receiving compartment does not require its downstream output to remain unchanged. Calibrate illumination or actuator effects and the resulting dendrite-to-output consequence separately; ex-vivo selectivity does not automatically transfer to behaving animals. Analyze animals, sessions, branches and trials at their declared levels, rather than treating recorded branches as independent animals.

A timing claim requires an adequately verified intervention with compatible, disrupted, matched-control and restoration conditions where feasible. Distinguish assigned scheduling policy from achieved phase and timeout selection. Human EEG/MEG cannot establish selective dendritic intervention; a circuit assay and a human access assay may require different preparations. This is a prospective research design requiring qualified investigators and ethics review, not a stimulation procedure.

Calibration must distinguish intended timing, uncertainty about achieved physiological phase, and the recording of the resulting response. Phase-estimation reliability can vary with signal quality and momentary amplitude; agreement with an offline estimator does not make latent physiological phase known [50]. A learning-related change in the accessible signal could consequently change targeting fidelity as well as neural function. Freeze the estimator and eligibility rules before confirmatory outcomes, assess delivery bias and uncertainty across training and retention, and report failures. Do not remove a genuine learned state change by silently conditioning the primary effect on post-training amplitude or accurate targeting.

Control stimuli can also change the nervous system rather than only its recording. TMS-EEG studies

illustrate why sensory co-stimulation and an instrument artifact require different interpretations: real neural responses to peripheral inputs can accompany or interact with the response of interest [51,52]. This comparison does not select TMS for the proposed touch experiment. Matching reported sensation or subtracting a sham response does not, without further assumptions, isolate an unchanged receiving process. Controls and calibration must be justified in the actual preparation; this manuscript supplies no transferable numerical bound.

Validate an intervention's selectivity in the preparation and state used to test the claim. A control from a different cortical area, age or anesthetic condition requires an explicit transfer argument. Report absolute responses for credited and comparison inputs alongside any preference ratio; test their differential change directly. Retain unpaired and no-postsynaptic-pairing comparisons where the circuit assay permits them. Randomization should be accompanied by masked preprocessing, exclusion decisions and outcome scoring where feasible; predeclare unavoidable unblinding, freeze outlier rules and report an all-eligible sensitivity analysis. These safeguards add no acquisition arm and do not require misleading participants about compensation.

Randomize the probe policy before assessing eligibility, and retain assigned trials for its primary effect. Phase-targeted and phase-randomized policies should share a prespecified eligibility rule; a replayed or wall-clock schedule answers a different control question. Eligibility can itself change after acquisition, so restricting the learning comparison to eligible bursts can select different populations. Later events in a closed-loop sequence can depend on earlier stimulation: equal rules or mean rates do not guarantee equal exposure. Do not manufacture a phase-only contrast by conditioning on those treatment-induced differences.

The timing policy must also define what happens to the tactile probe when delivery fails. Record actual sensory presentation, intervention delivery, usable neural recording and available response separately. A timeout is not an intentional catch trial. Where feasible, retain the tactile probe and behavioral assay under a prespecified fallback; log its input and timing. The all-assigned effect still includes fallback and waiting consequences. If a probe must be aborted, report availability and the declared policy composite separately from the sensory endpoint. Appendix I shows that changed presentation probability alone can create both timing interactions with unchanged delivered-event responses. Neither discarding failures nor silently counting them as ordinary misses isolates phase-sensitive reception.

The decisive mechanistic comparison also separates one useful receiving window from any additional benefit of repeated temporal organization. Compare an isolated-event account with regular and irregular histories while preserving the immediate test relationship and declared input budgets as far as feasible. One event and many cannot simply be called dose-matched. Recovery, actuator dynamics, propagation delay and receiving-state differences remain explicit alternatives. If an isolated event reproduces the benefit within justified precision, a necessity claim for repeated cycles weakens; a residual history-sensitive effect still needs its own causal explanation.

Measure hits, false alarms, feature or interval discrimination, perceived magnitude, confidence and access reports separately. Counterbalance response mappings and comparison order. A criterion change can reflect altered subjective evidence or decision strategy; forced choice, ratings and reduced-report measures each add constraints without becoming direct experience meters. Do not condition the main analysis on correct or reported-aware trials. Preserve valid behavior when only neural recordings fail.

Keeping randomized assignments does not recover an unobserved endpoint. Report counts and reasons for presentation, intervention, recording and response failure in each prespecified cell. Preserve complete behavior when only recording fails. A declared missing-data model or sensitivity analysis is still needed; Appendix I provides sharp completion bounds for finite-cell binary means and their linear contrasts without treating missing responses as zero. These are descriptive bounds, not population confidence intervals or

bounds for unbounded neural quantities. Distinguish inconclusive observation from failed learning, failed expression and verified preservation of function.

An executable model comparison must distinguish the time a state represents from the time its estimate was computed [55]. The qualified solver revises its modeled trajectory as new evidence arrives; a returned column about an earlier step need not equal the belief available at that step. Record online snapshots for time-specific neural comparisons and use the action distribution saved at the actual decision for response scoring. Later teaching feedback can change a final policy weight after the action has already occurred. Task identity must accompany the readout: left/right choices and seen/unseen responses use different meanings even when their numerical codes or array dimensions agree. Appendix L gives the executed timing contrast and exact action mapping.

6.3 Six predictions and informative failures

ID	Prediction	Result that would constrain the claim
P1: contingency	At matched probe input, contingent learning produces more feature-specific transfer or recruitment change than replay and active noncontingency.	Precise equivalence weakens the added-contingency claim; shared improvement favors exposure or practice.
P2: relative influence	Learning changes the prespecified phase-compatible source-receiver relation and its downstream consequence, with explanatory value beyond input, rate/power, and measured state.	No incremental held-out value weakens this phase-mediated formulation. An association alone does not identify a causal route.
P3: timing intervention	Verified disruption preferentially reduces the learned advantage, and compatible restoration recovers it more than control timing.	Timing-independent impairment favors nonspecific disruption; preserved benefit after effective selective perturbation challenges the specified causal step.
P4: conscious access	The supported training-by-timing effect also changes prespecified access outcomes after sensitivity, criterion, confidence, and reporting alternatives are addressed.	Changes confined to performance, reaction time, or task-bound reporting do not establish the added access relation.
P5: reciprocity	Learned expectation changes tactile recruitment, while tactile evidence changes the expectation's later influence.	A verified exclusively one-way effect constrains the proposed reciprocal route. Missing measurements do not establish either direction.
P6: value context	With tactile input controlled, learned appetitive/aversive context changes receiver-relative influence and behavior.	A complete explanation by measured arousal, criterion, or generic rate changes removes the claimed incremental mechanism. No universal voltage sign follows.

These predictions preserve different responsibilities. P1 concerns acquisition under a control probe. P3-P4 concern whether that acquisition changes a timing-dependent advantage and its relationship to access. Both are required for the stronger joined claim, but a learning benefit independent of timing may support the broader learned-recruitment account while weakening this phase-specific version.

The prospective protocol estimates contingent-versus-replay and contingent-versus-active-control contrasts separately. Averaging controls can hide equality to informative replay. A superiority-to-both claim needs both prespecified contrasts; its conjunction logic does not supply valid component tests, multiplicity correction for every endpoint or a biological power calculation. Participant, yoke-set, recording and trial levels must remain distinct. Weak-null inference also requires assumptions beyond those of an exact sharp-null randomization test [30].

Randomizing acquisition and a timing intervention does not, by itself, identify mediation through a recorded neural variable. The causal route needs additional assumptions and intervention specificity. Failure to achieve the intended perturbation is inconclusive, while precise preservation of the learned advantage after an effective selective disruption challenges the specified mechanism.

For P2-P4, positive contingent-versus-replay and contingent-versus-noncontingent timing interactions do not alone establish increased intrinsic receiving modulation. The phase-delivery counterexample yields both interactions with an unchanged response-versus-true-phase function. Preserve the total randomized scheduling-policy effect, then assess the narrower mechanistic claim under a justified delivered-phase/reference model. If only delivery improves, that supports a different explanation of the policy benefit; it does not establish absent learning. If calibrated receiving change also predicts and causally affects the declared access outcome, the connected proposal gains more specific support. Neither result identifies the constitutive experience claim by itself.

For the stronger P2-P4 connection, the specified learned receiving relationship should predict a downstream consequence and the access effect of a validated disruption or restoration under the declared observation rules. A change confined to event locking, task retention or reporting supports that narrower operation. It is not interchangeable with causal use of the proposed interpopulation relation.

Full observation equations, contrast definitions, independent-calibration requirements and failure policies remain in the supplement and prospective protocol, read with its acquisition/expression amendment and timing-specificity amendment, followed by the separate feedback-credit extension and its credit-specificity clarification. The design is unregistered and unexecuted. Pilot reliability, feasible measurements, effect-size targets, enrollment planning and specialist statistical/ethics review remain necessary before biological testing.

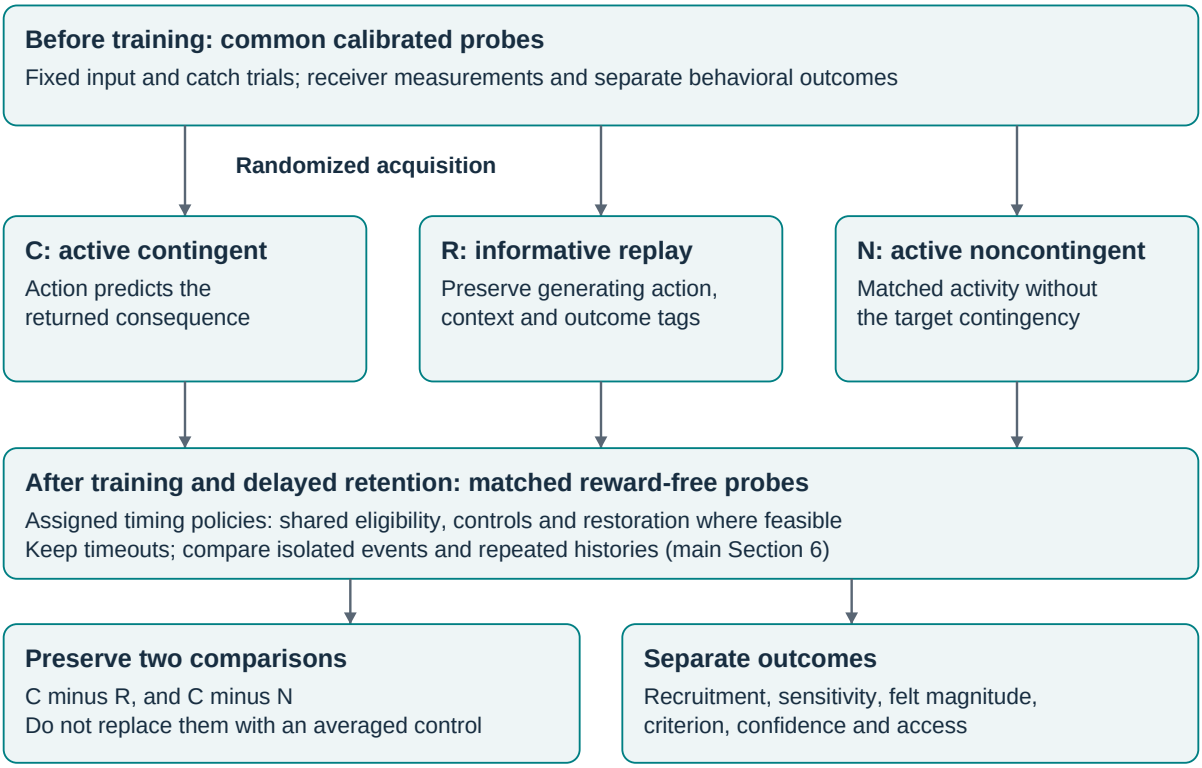


Figure 2: Acquisition-to-access design. Reward-free probes and two separate control comparisons distinguish acquired change from current reward and exposure. Timing policies retain timeouts and use declared eligibility rules; isolated-event and repeated-history comparisons test different mechanisms. Context reinstatement is a separate, prespecified extension, not pictured as another acquisition arm. Access measures remain distinct from neural or behavioral performance. The feedback-credit follow-up is separate and is not pictured. Cumulative design refinements appear in Section 6 and the technical supplement appendices; the optional stage-specific circuit companion is not pictured. This is prospective and unregistered, not an executed experiment.

7. Constructed applications: what works, and what the alternatives explain

7.1 Design and inference checks

Application 01 uses exact constructed response functions to expose confusion among gain, retuning, response scale and reference drift. Its 14 checks establish software/algebra behavior, not sensory learning. Application 02 simulates receiver-level estimation with heterogeneous gains and independent coefficient estimates. Under its assumptions, an imposed rotation becomes easier to detect as the effect or sample grows; those conditional results are not biological power or enrollment recommendations. Every condition, rejection count and uncertainty estimate is retained in the supplement and result files.

The separate 13 access-measurement fixtures expose nonidentifiability of sensory evidence versus threshold, distortion from pooling participants, and failure of averaged controls. They are deterministic examples, not another learning experiment.

7.2 Learning changes consequential temporal selection

Application 03 learns a readout from transient sensory history, a familiar computational principle [31]. Twelve fixed seeds per synthetic environment, 384 training episodes and 256 held-out tests compare integration, phase features and independent time-bin weights. All learners receive the same action/context information and supervised-equivalent feedback; they are not parameter-capacity matched.

Constructed environment	Integrated receiver	Phase-feature receiver	Time-bin receiver
Both pulses informative	96.78%	96.55%	97.88%
Timing distinguishes target from distractor	73.76%	91.80%	93.42%

The phase-feature receiver learns selective influence when timing matters, but the nonoscillatory time-bin learner matches or exceeds its accuracy on every paired seed. In the other environment, the phase receiver has no observed advantage over integration. Both are results, not conditions selected after seeing which favored the hypothesis.

Informative replay exactly reproduces active learning; degrading the action tag instead tests lost information. Misregistering the temporal reference impairs both temporal receivers, not only the phase model. A restoration therefore demonstrates a software relation, not endogenous neural oscillation. The initial weights are zero, so acquired selectivity is not rotation of an already defined preference.

The open-loop replay in the mouse experiment [53] uncoupled current locomotion from visual flow. It is not the same intervention as this learner's information-preserving replay, which supplies action/context labels and equivalent teaching information. Their different outcomes make the control's retained information and bodily state part of the explanation; neither establishes that action execution is universally necessary or dispensable for sensory learning.

Retained weights survive reset of episode state, and reversal learning changes the interpretation of fixed held-out input. Disconnecting the decision makes learned content decodable to an analyst but unused by the system. Together these operations implement a learning-to-influence subchain, not learned consciousness or the full recurrent reconstruction.

7.3 Learned reconstruction and action for additional evidence

Application 04 estimates a joint Gaussian model from complete training vectors over predefined cues and a consequence. Learned covariance supports reconstruction from partial noisy observations; a decision uses the estimated consequence. Recurrent prediction has substantial prior foundations [32]. Here training is global statistical estimation, not a local plasticity rule or an executed action/reward training loop.

Twelve constructed instances share 512 training episodes and 128 held-out tests across conditions. Targeted and random observation acquire one extra cue, never the hidden consequence. Values are equally weighted seed means. The common reconstruction endpoint uses the same six initially unobserved coordinates, including any later sampled coordinate.

Condition	Decision accuracy	Consequence MSE	Common six-cue MSE
reciprocal	77.41%	0.7540	0.6562
direct	77.41%	0.7540	0.6562
block_lesion	65.69%	1.2062	0.9130
marginal_block_control	69.73%	0.9966	0.8013
shuffled_target	58.33%	1.4729	0.6562
reused_observations_x4	77.21%	0.7856	0.6707
greedy_extra	83.72%	0.4287	0.4686
random_extra	79.69%	0.5935	0.4629

The recurrent solver converges in every tested inference, and direct solving matches every initial decision. Targeted observation reduces consequence error versus random observation on every paired seed, but does not improve every seed's accuracy or the mean common-cue reconstruction error. Selecting useful evidence for an action is different from globally maximizing detail.

Two post-result amendments remain disclosed and reproducible. The original precision-block lesion changed uncertainty as well as interaction; a marginal-covariance-preserving control was added. A second amendment made reconstruction comparisons use common coordinates. Earlier metrics, code and results were retained rather than replaced.

Shuffling learned cue-consequence relations damages task use while leaving cue reconstruction effectively unchanged. Reusing observations as independent unjustifiably narrows model uncertainty: consequence error and log loss worsen, but Brier score slightly improves. High-confidence error fractions remain below 10%, so these results do not warrant claiming that every calibration measure deteriorates. Every endpoint and denominator is in the supplement.

These models establish possible functional joins and expose alternatives. They do not form one trained end-to-end SAN system. Their contexts, feature coordinates and temporal reference are supplied; they do not establish raw representation learning, phase-specific biological reception, conscious access or a necessary recurrent implementation. The limits accompany the constructive results rather than cancel them.

7.4 Joint observations distinguish some stages, not every mechanism

Application 05 addresses the missing connection between a changed receiver and changed behavior. A receiver readout acquires weights from 384 synthetic training episodes under Application 03's update rule. The weights are then fixed. Two supplied perturbation families alter gain or temporal registration, while matched probes and a noisy receiver observation support three candidate explanations: receiver change expressed in behavior, a later policy change with the measured receiver unchanged, and separate

receiver/policy changes. The policy alternative retains the full temporal input and intervention label; it is not denied the information needed to perform the task.

Twelve fixed seeds, four generating cases per perturbation family and three candidates give 288 fits. Generators include receiver, policy, mixed and null cases. Baseline response calibration and intervention fitting use separate data; held-out perturbation levels and stimulus-amplitude conditions receive no fitting. The mixed candidate has one additional parameter. Receiver scale/noise and the observation equations are stipulated, not measured physiological properties. The binary outcome is a synthetic response, not a conscious-access measure.

The table reports mean joint negative log score per observation across seeds, combining receiver and response prediction in fixed declared units. Lower is better.

Perturbation	Generating case	Receiver-only	Policy-only	Mixed
Gain	Receiver change	0.709099	1.827781	0.709888
Gain	Policy change	0.726905	0.709087	0.709888
Gain	Mixed change	0.752451	1.863318	0.745425
Gain	Null	0.773184	0.773605	0.774407
Registration	Receiver change	0.809354	2.595938	0.810187
Registration	Policy change	0.987949	0.808202	0.810187
Registration	Mixed change	0.881890	2.591079	0.805328
Registration	Null	0.767794	0.769009	0.770994

Receiver and policy generators produced identical response probabilities and paired responses, and their behavior-only fitted families were identical. With joint observation, the correct restricted model beat the opposite restricted model on every paired seed in those two generators. The policy-minus-receiver mean score difference was 1.118682 for receiver-generated gain and -0.017819 for policy-generated gain; corresponding registration differences were 1.786583 and -0.179748. Their Monte Carlo standard errors were 0.033374, 0.001310, 0.077114 and 0.008333. These quantify synthetic seed variation, not participant uncertainty, biological significance or power.

Mixed generation exposes a limitation of the shared-coefficient explanation. The mixed candidate improved on receiver-only by 0.007026 for gain and 0.076562 for registration on average, but the gain advantage failed on one seed. The null-case restricted-model differences straddled zero across seeds. Small fitted effects under this known null are not an empirical equivalence test. Every condition and endpoint remains available, including imperfect coefficient recovery.

The mapped nonoscillatory implementation matched all joint predictive metrics to within 1.11e-15. It re-expresses the acquired weights as temporal coefficients, rather than independently learning a larger model. Thus, the added observation helps identify where a change appears within these constrained models without identifying an endogenous oscillator. Nor does it prove mediation: an upstream cause can generate the measured receiver change and a separate decision effect. Appendix E supplies that counterexample and the explicit measurement equations.

This construction advances the proposed learning-to-influence connection by specifying what an additional measurement buys and what intervention remains necessary. It does not complete the acquisition-to-experience chain. The frozen protocol, implementation and complete numerical results preserve the design and all outcomes; Appendix E gives the integrated technical account.

Appendix F expands the observation assumption analytically after these results. Its recording-alias example and calibration bounds leave all 288 fits unchanged. The accompanying deterministic identity checks are not another learned application, an estimate of measurement reliability, or a biological power calculation.

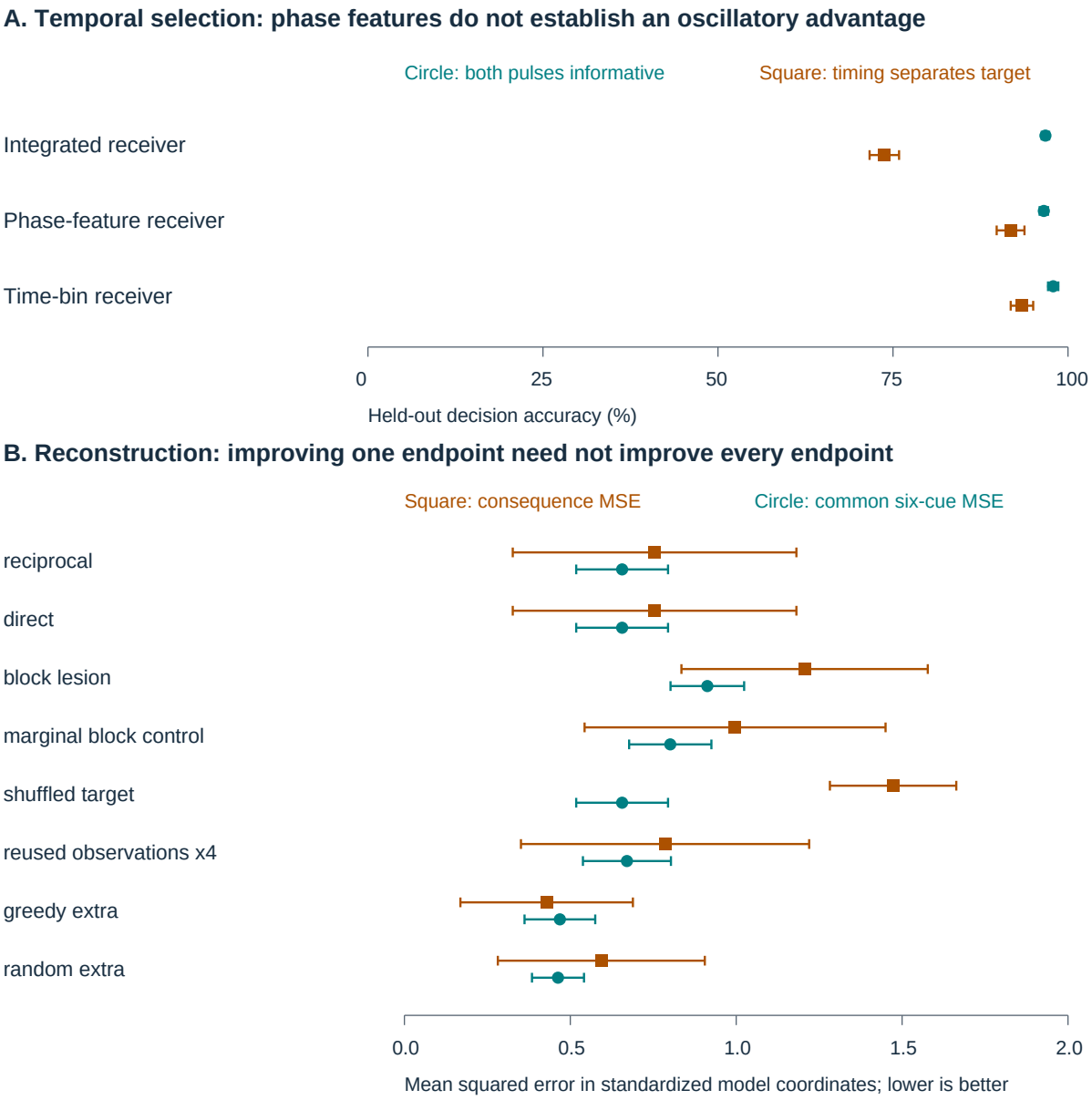


Figure 3: Constructed results and alternatives. Panel A shows six intact contingent-training summaries from Application 03; informative tagged replay matches them exactly in the saved fits. Panel B shows all eight Application 04 conditions for consequence and common six-cue mean squared error. Points are seed means and whiskers are one sample standard deviation across 12 synthetic seeds per condition, not biological uncertainty or confidence intervals. Direct inference reproduces recurrent reconstruction. Target-directed observation improves consequence prediction but does not outperform random observation on every endpoint. Both post-result amendments remain declared in the text and supplement. No models were refitted for this figure.

7.5 An executed external comparator with distinct feedback and task roles

A separate qualification uses the published six-step presence-report model [55], its exact supplied solver and a declared SPM12 dependency revision. The authors' exact dependency revision remains unknown. Four fixed source-setting cases establish runtime behavior, and an upper-register handoff separates an acquired parameter snapshot from fresh episode state. This is not a sixth fitted application or an implementation of the full active-inference theory.

For this qualification, one fixed-seed acquisition episode in the stronger source setting supplies the saved counts, and all six probes retain that source setting. This is not a general acquisition trajectory or an estimate of learning across stimuli.

Two task-preparation contexts then use the same saved upper-level likelihood, transition and initial-state counts. Each context has native, zero-mask adapted and announced-feedback-withholding probes. Native and zero-mask outputs match exactly across twelve checked fields, including state/policy beliefs, sampled outcomes, counts and utility. The withholding adaptation removes received terminal correctness evidence and anticipated information from that channel while preserving the utility of correct responding.

Task preparation	Probe condition	Internal report state	Generated correctness code	Correctness feedback received
Report	Native	3	3	Yes
Report	Zero mask control	3	3	Yes
Report	Announced withholding	2	2	No
No report	Native	2	2	Yes
No report	Zero mask control	2	2	Yes
No report	Announced withholding	2	2	No

Internal report codes are null, unseen and seen; correctness codes are null, incorrect and correct. The no-report rows are not administered awareness responses. These single fixed-seed outcomes supply no detection rate, uncertainty interval or learned advantage. The report-context change belongs to the complete announced-availability policy, not solely to deleting a late observation. The retained acquisition artifact remains fixed, so changed expression is not evidence that it was erased. Nor is either result a conscious-experience measure.

The complete runtime record preserves inputs, exact control checks, preceding failures, recovery and terminal process receipts. This closes a defined source/adapter qualification. Sections 7.6 and 7.7 extend the retention and common-task interfaces; held-out receiving interventions and calibrated receiver/access observations remain necessary for the substantive SAN comparison. No theory-wide winner is claimed.

7.6 Informative replay and retained learning at both levels

The published comparator [55] now supports a declared two-level retention rule. The source's upper likelihood, transition and initial-state counts are retained, together with the sensory-level likelihood counts. At the episode boundary, each sensory child's posterior count increment is added to a common retained template. Native inference within the episode remains unchanged. Appendix L specifies this batch rule, its count bound and the distinction between physical process fields and received evidence.

One native acquisition episode supplies a recorded process and action history. A fresh replay constructor receives that root history; the source's deterministic child process regenerates the same sensory observations.

Native and replay inference agree exactly across twelve recorded fields, and six child comparisons preserve their observations, beliefs and sensory counts. The replay uses a different random seed. Thus, under this specified interface, the informative replay preserves the acquired state at both levels. This strengthens the existing replay counterexample: the model's retained learning requires no additional execution variable once its usable history is supplied. Human agency, engagement and exposure remain further differences to investigate in the prospective study.

The six child updates add approximately six units of sensory count mass. The largest change in a normalized sensory likelihood entry is 0.01971. This is a change in the model's observation mapping; the count units and likelihood probabilities are not calibrated electrical magnitude or experienced intensity.

Six feedback-withheld probes then compare upper-only retention with retention at both levels. Both conditions use the same acquired upper counts, current task, process and seed. The trained left stimulus, untrained right stimulus and intentional blank catch each receive one probe in each condition. All returned probe changes are discarded, including the upper initial-state count updates that remain possible at zero scalar learning rate.

Presented stimulus	Maximum sensory posterior difference	Maximum final-trajectory posterior difference	Internal response code, upper-only / both levels
Trained left	0.001770	0.073258	2 / 2
Untrained right	0.004840	0.004497	2 / 2
Intentional blank catch	0.003847	0.000802	2 / 2

The differences are maxima over the recorded posterior entries, not distances between whole experiences. The sensory maximum uses the two stimulus-presentation steps. The upper maximum compares the final update's estimates for four state factors across all six modeled steps. It describes a final inferred trajectory; an online-state comparison requires the separate snapshots described in Appendix L. The source response code 2 denotes unseen. The largest sensory difference occurs for the untrained stimulus, so these observations do not establish feature-selective improvement. Retaining sensory learning changes internal inference without changing the selected report in these fixed cases. Both signal presentations produce unseen responses, while the catch's unseen response is correct for presence discrimination. The common response code therefore hides different task outcomes.

These are executed comparison-interface results from one acquisition episode and one probe per cell. They supply neither an acquisition learning curve nor a population detection estimate. Section 7.7 adds finite contingent, replay and noncontingent acquisition histories. Repeated held-out tests, declared receiving interventions and calibrated observation/access measures remain further requirements. The capable conventional learner and the SAN-specific receiving proposal must receive the same usable history. These results strengthen that comparison while leaving the proposed causal phase mechanism and constitutive experience claim open to their further tests.

The case records preserve source bindings, parameter snapshots, all eight completed inversions in the accepted cases, the first rejected replay attempt and the read-only checks. The five constructed manuscript applications remain unchanged.

7.7 A finite common task and its calibration limit

A further computational check places the three existing linear feature families and two settings of the source-derived orientation model [55] on a common 32-episode task. A prescribed action controls the diagnostic return in the contingent role; an independent source controls it in the active noncontingent role. Both roles have the same sensory marginals. Informative replay supplies the contingent observations with their truthful context and action tags. The receiving transform and exploratory policy are provided, not learned. Appendix M specifies the construction and its conditional results.

After acquisition, the phase-feature and ordinary time-bin receivers each correctly classify all four intact probe patterns in the contingent and informative-replay roles. The integrated receiver classifies three of four. That difference has a precise representational explanation: its summed input merges two patterns with opposing targets, whereas both finer feature families can recover the first pulse. It is not an exclusive oscillatory achievement. In the active noncontingent role, the corresponding accuracies are one half, one quarter and zero for the integrated, phase-feature and time-bin rules. Those extreme thresholded outcomes belong to this particular structured history; they are not population estimates or a general inverse-perception effect. The full score errors and input-influence coefficients are retained.

The source-derived comparison observes its own response and correctness during acquisition. Its update-free probes retain own-response knowledge while withholding correctness. Two predeclared settings scale retained counts by one and one eighth. They share initial normalized count mappings but are not capacity- or learning-rate-matched to the linear learners. At 32 episodes, both settings and both executed acquisition roles have mean correct-action probabilities within 0.000007 of one half. Their retained parameters change, but this configuration does not acquire accurate discrimination. It therefore does not yet supply a task-calibrated competitive baseline, and its weak performance cannot support a claimed SAN or phase-coding victory.

The declared finite workload is complete: 172 source-model inversions, nine linear fits and direct inspection of all 128 retained acquisition states. Source-model informative replay is an explicitly derived identical-input alias with coupled randomness, not another independent run. The linear replay calculations are separately executed and identical to their contingent counterparts. Probes use the acquisition input alphabet; they do not demonstrate unseen-stimulus generalization. This completed interface test leaves the broader comparison open to adequate source-model calibration, multiple histories, held-out receiving interventions and physiological/access measurements. Left/right choices in this task are not seen/unseen reports or measures of experience.

8. What the proposal says about experience

In the SAN account developed here, the meaningful sensory event is not just a local input plus an externally assigned label. Its meaning is expressed through the learned relationships it recruits, the predictions and possible actions those relations support, and their continuing revision through returned evidence. The functional connections are physically realized, even though no single scalar electrical quantity captures their significance.

I further propose that an appropriately organized recurrent self-world learning process constitutes the experiencing system. This is stronger than asserting that a circuit correlates with a report, and it remains a hypothesis. The present mathematics does not derive phenomenology from phase. Nor does a positive behavioral experiment prove that every realization of the measured function has experience.

The scientific task is nevertheless productive: identify which relations remain available, which influence the system's own subsequent operations, and which are necessary for the declared access condition. Those results can progressively constrain the larger account without disguising its remaining explanatory burden.

9. Scope, limitations and conclusion

The contribution is a specified acquisition-to-access hypothesis, a set of conditional identification results, constructed functional examples and an experimental program. It is not a new biological finding or a demonstrated theory-wide advantage. The evidence connects compatible operations across different preparations; the joined causal mechanism remains to be established in a specified sensory relation.

The examples explain how retained learning can change later use, but favorable assumptions remain visible. They use supplied features and references, simplified feedback and Gaussian reconstruction. Their strong alternatives show that task success alone cannot identify oscillation, bodily agency or recurrent realization. Physiological timing, learning rules and report generation remain research targets rather than properties silently inherited by the models.

The source review is targeted, not systematic. Some studies received selected methods/Results review; others remain abstract-level or source-access limited. Their reference presence is not a claim of complete verification. Historical SAN notes document development without converting owner-held dates into independently verified early public priority. My earlier theoretical papers provide foundations, not independent experimental confirmations of this extension.

The positive proposal is that learning can change both what a sensation means and how it participates in the network. Its significance can be physically realized through acquired relationships, selective reception, inhibition, excitation and recurrent use - not a universal correspondence between emotional sign and voltage. This makes a concrete empirical question: with input held comparable, which learned change in receiving organization alters what the organism can feel and do next?

Answering that question requires testing the connections together. A useful result could support learned access while rejecting a particular phase mechanism, or establish receiver-dependent processing without establishing conscious access. The constitutive SAN claim remains explicit: an appropriate self-world learning organization is proposed to be the experiencing system. The present derivations constrain its functional investigation; they do not settle phenomenology.

Reproducibility and research status

The original eleven conditional proofs, complete constructed results, adverse comparisons and methodological amendments are retained in the technical supplement and its linked artifacts. No biological dataset was generated or reanalyzed for this manuscript. The empirical protocol is prospective, unregistered and subject to feasibility, statistical and ethics review. Clinical benefit and conscious AI are not established.

The additional joint-measurement application and three conditional identification results are presented in Appendix E. They extend the original technical material without replacing its equations, proofs, numerical results or amendments. Joint predictive success, causal mediation, oscillatory implementation and conscious experience remain different claims.

The executed comparator qualification and Appendix K preserve the original source and the declared derivative, parameter snapshots, six successful probe conditions, failed attempts and read-only verification. Appendix M adds the completed finite common task, its elementary conditional results and the comparison release note. These computations do not establish new biological data, a calibrated competitive theory comparison or independent scientific approval. Elapsed computer time is not a calibrated neural or perceptual latency.

AI assistance was used in source exploration, drafting, mathematical analysis and code development. This author-authorized preprint is not independently peer reviewed. Version 1.0 releases selected original code; third-party comparator code and the complete private replay package are excluded pending redistribution clearance. No study enrollment is claimed.

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